

练习三 力学基本定律（三）

1. $20\text{rad}\cdot\text{s}^{-2}$; $48\text{rad}\cdot\text{s}^{-1}$

2. $\frac{4}{3}\omega_0$; $\frac{1}{2}J\omega_0^2$

3. (3); 4. (2)

$$mg - T = ma$$

5. $TR = \frac{1}{2}MR^2\alpha$ $\alpha = \frac{mg}{(m + M/2)R}$; $a = \frac{mg}{m + M/2}$;
 $a = R\alpha$

6. (1) 由角动量守恒得: $J_1\omega_1 + J_2\omega_2 = 0$

$$MR^2 \cdot \frac{v}{R} + J_2\omega_2 = 0 \quad \omega_2 = -\frac{mRv}{J_2} = -0.05(\text{S}^{-1})$$

(2) $[\omega_1 - (-\omega_2)]t = 2\pi$ $t = \frac{2\pi}{0.55} \text{ (s)}$ $\theta = \omega_2 t = \frac{2\pi}{11} \text{ (rad)}$

(3) $T = \frac{2\pi}{\omega} = \frac{2\pi R}{v} = 4\pi \text{ (s)}$ $\therefore \theta = \omega_2 T = 0.2\pi \text{ (rad)}$